SPENT UP TO **15 HOURS** A DAY RECORDING HIS **OBSERVATIONS**

PUBLISHED AROUND **100 ARTICLES** AND MADE MORE THAN **2,900 DRAWINGS**

SPAIN'S **CAJAL INSTITUTE** HOLDS **30,218** ITEMS IN **"CAJAL LEGACY"**

Camillo Golgi's method (see box) and was amazed. It was a career-changing moment for him. "The nerve endings could be seen," he wrote later. "A look was enough. Dumbfounded, I could not take my eye from the microscope."

After trying Golgi's method, Cajal worked to improve it until it allowed for the complete visibility of the nerve cells in the brain, eye, and spinal cord tissue of birds and embryonic mammals. He then drew by hand, in intricate and exquisite detail, the stained cells he saw through his microscope, creating some of the world's greatest scientific illustrations.

Neuron doctrine

Over a 6-year period, Cajal continued to apply his modified staining technique to other kinds of nervous tissue from different animals, including humans. By 1889, he had observed and recorded that the brain and nervous system are composed of billions of individual, independent nerve cells that communicate with each other—through electrical and chemical transmission—across tiny gaps (now called synapses). Cajal's observations refuted the prevailing theory on the composition of the nervous system—reticular theory—which claimed that the nerve fibers were fused together to form a single, continuous network.

In 1891, German anatomist Wilhelm Waldeyer coined the term "neuron" to mean nerve cell and used Cajal's findings to support a "neuron doctrine"; this states that the basic units of the nervous system are discrete cells (neurons) that have a very specific connectivity that determines how they